

VHDL#1

ECSE 222 - VHDL Assignment 1

Adam Gangat | 261224674

Houssein Mourad | 261226763

Sanad AlAghbar | 261210889

PS: Unfortunately, we were unable to take screenshots on the lab machine owing to a technical issue, so we couldn't include any.

Executive Summary:

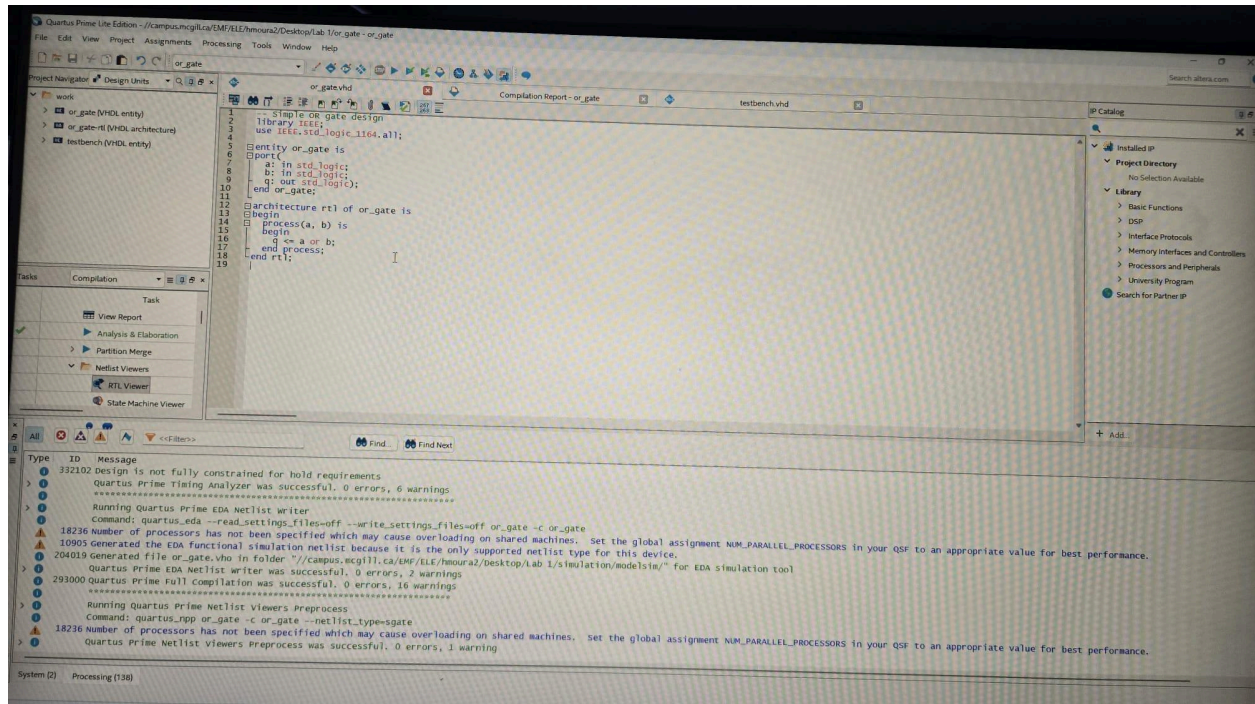
In this VHDL assignment, we were introduced to the Quartus Prime Lite design software for writing VHDL code and then verifying it using ModelSim. This was applied to two separate tasks.

The first task involved implementing an OR gate in VHDL, along with a corresponding testbench to verify whether the test code was correct. Furthermore, we then generated the associated circuit and simulation on ModelSim, where the waveforms represented the OR gate as expected.

Next, we applied the skills learned from the first task and implemented them for the XNOR gate, again verifying the correctness of the code through the same process and producing the corresponding circuit and simulation as seen below in the screenshots.

Results Obtained & Explanation:

Part 1: Or_Gate



Or Gate: VHDL Source

```
library IEEE;

use IEEE.std_logic_1164.all;

entity or_gate is

port(

    a: in std_logic;

    b: in std_logic;

    q: out std_logic);

end or_gate;
```

architecture rtl of or_gate is

begin

process(a, b) is

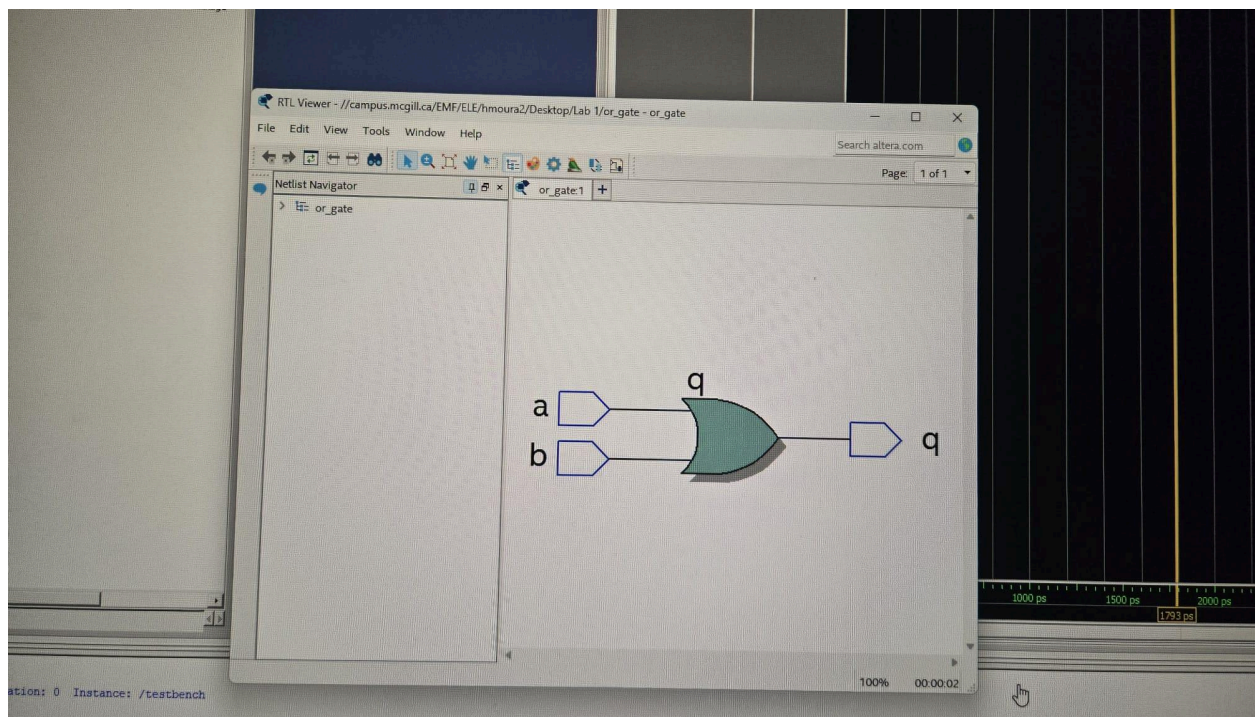
begin

q <= a or b;

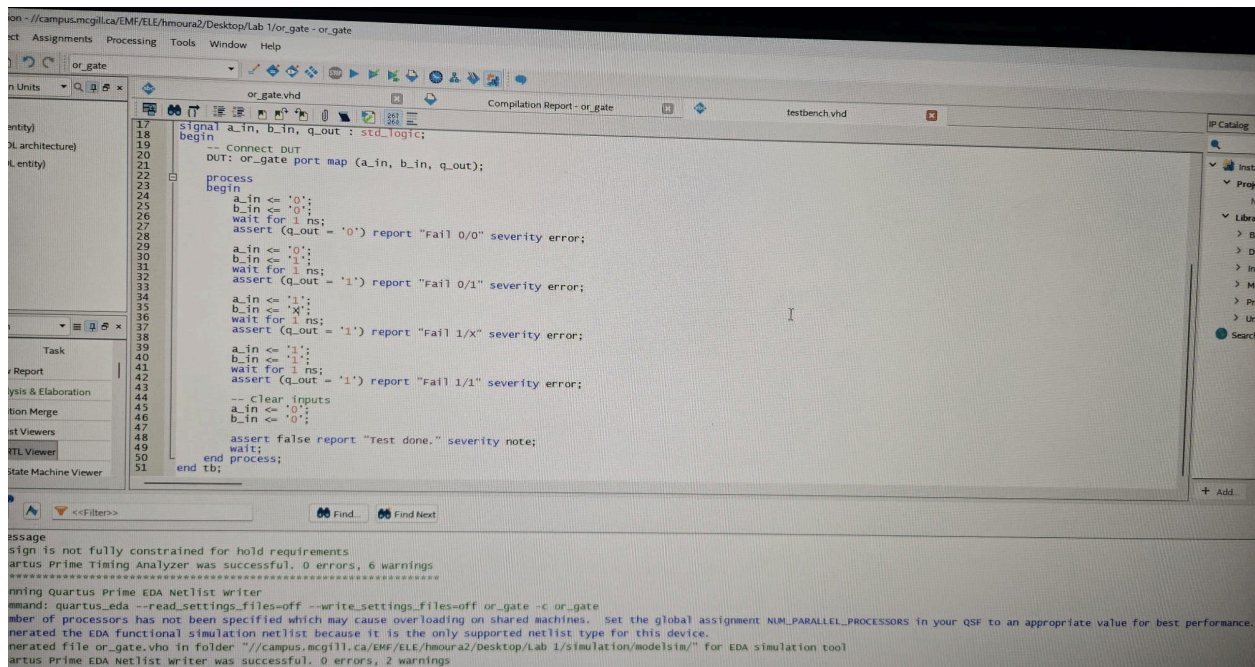
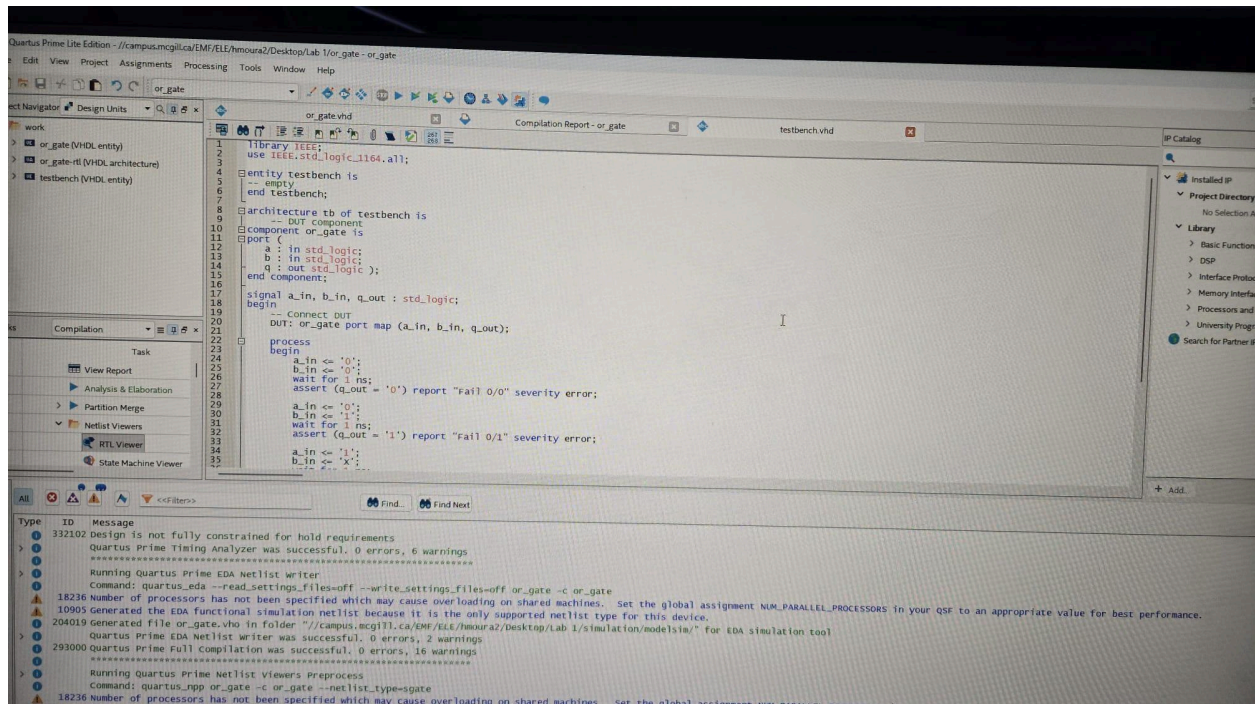
end process;

end rtl;

Shows the entity/architecture for a 2-input OR

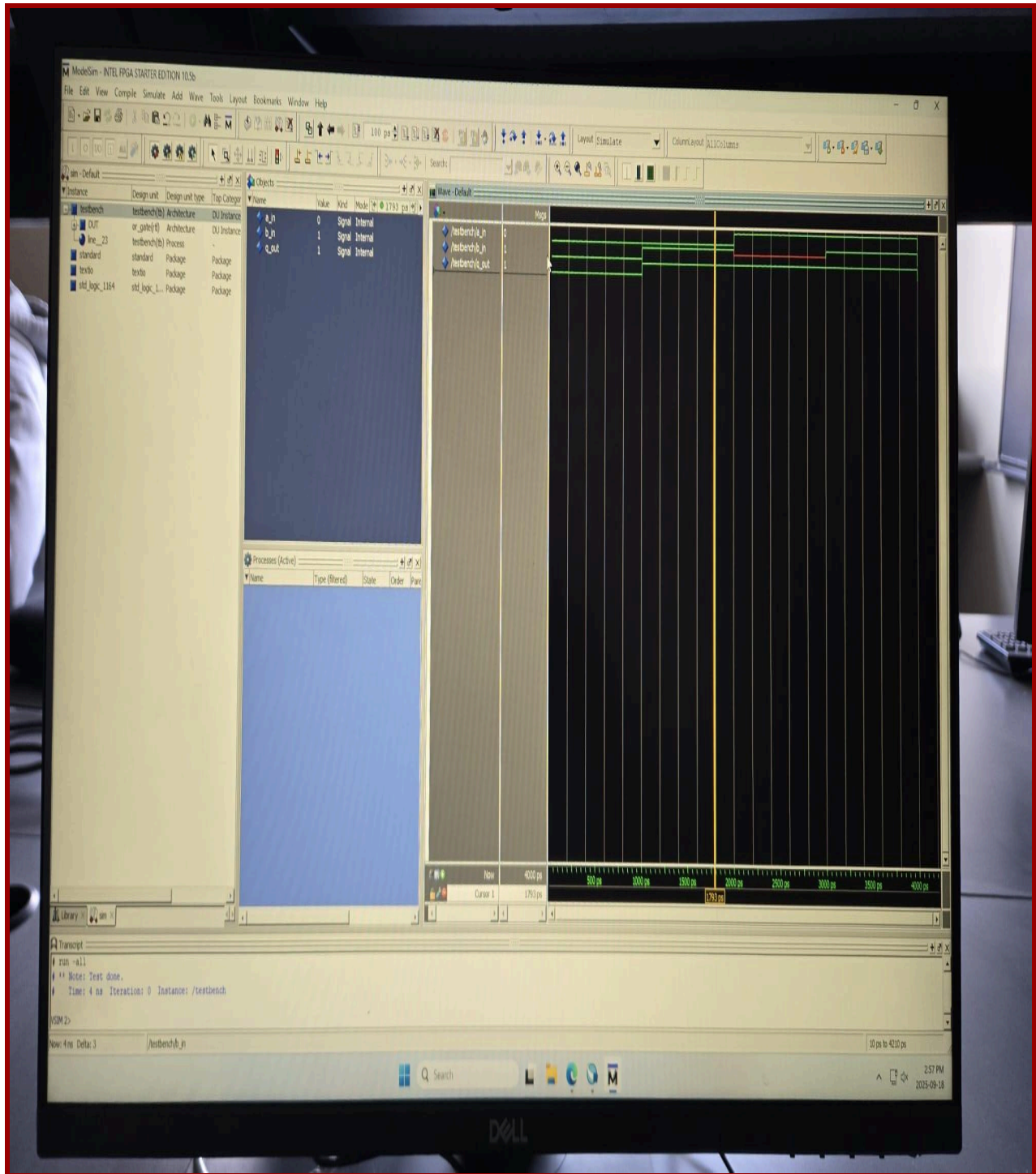


Logic synthesis result: Quartus implements the design as a single 2-input OR gate



OR gate test-bench:

This verifies the truth table and the 1-dominance behavior for 1X



The simulation result shows that the output q is 1 whenever either input a or input b is 1 (or both), and 0 only when both inputs are 0, which corresponds exactly to the behavior of an OR gate.

First variable	a = 0	a = 0	a = 1	a = 1
Second variable	b = 0	b = 1	b = 0	b = 1
Result	q = 0	q = 1	q = 1	q = 0

OR Gate simulation

Part 2: Xnor_Gate

XNOR gate design

library IEEE;

use IEEE.std_logic_1164.all;

entity xnor_gate is

port(

a: in std_logic;

b: in std_logic;

q: out std_logic);

end xnor_gate;

architecture rtl of xnor_gate is

begin

process(a, b) is

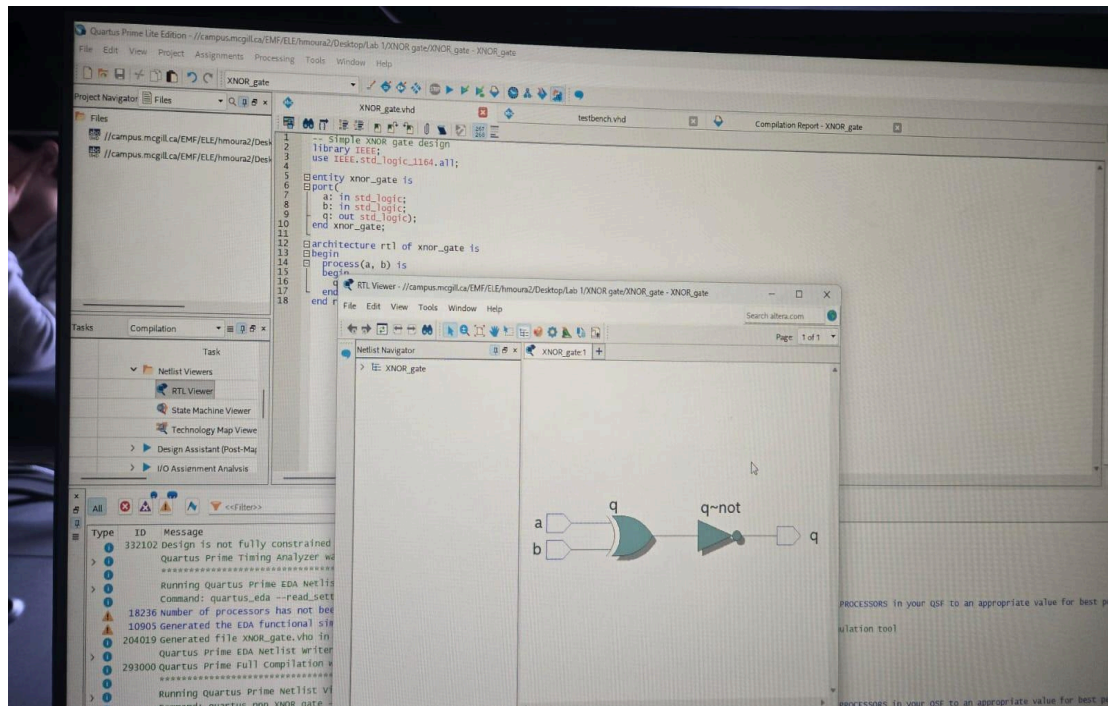
begin

q <= (a xnor b);

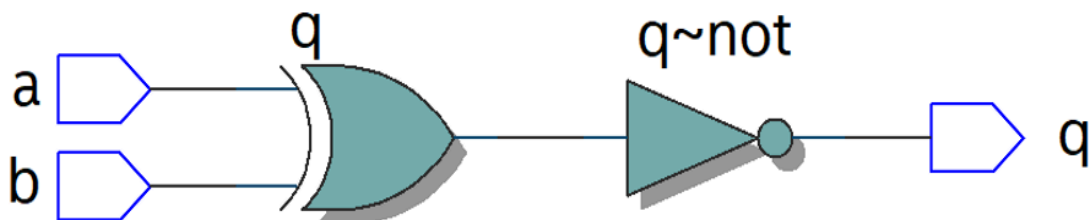
end process;

end rtl;

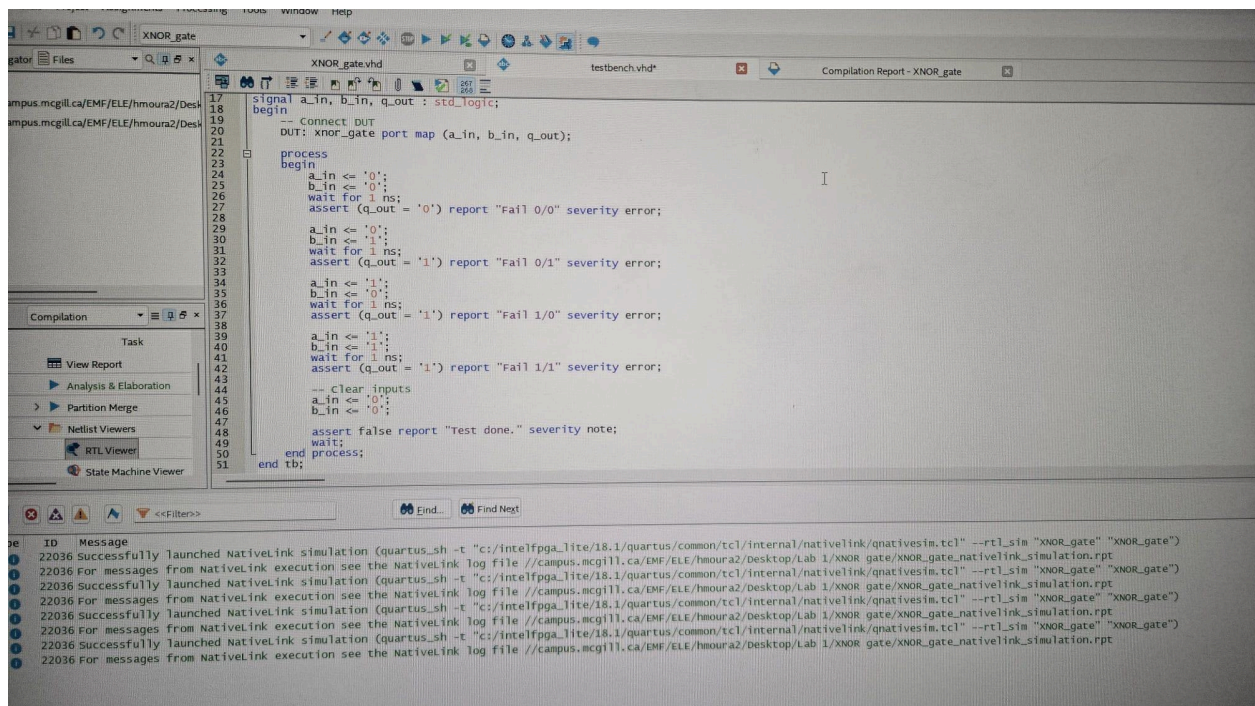
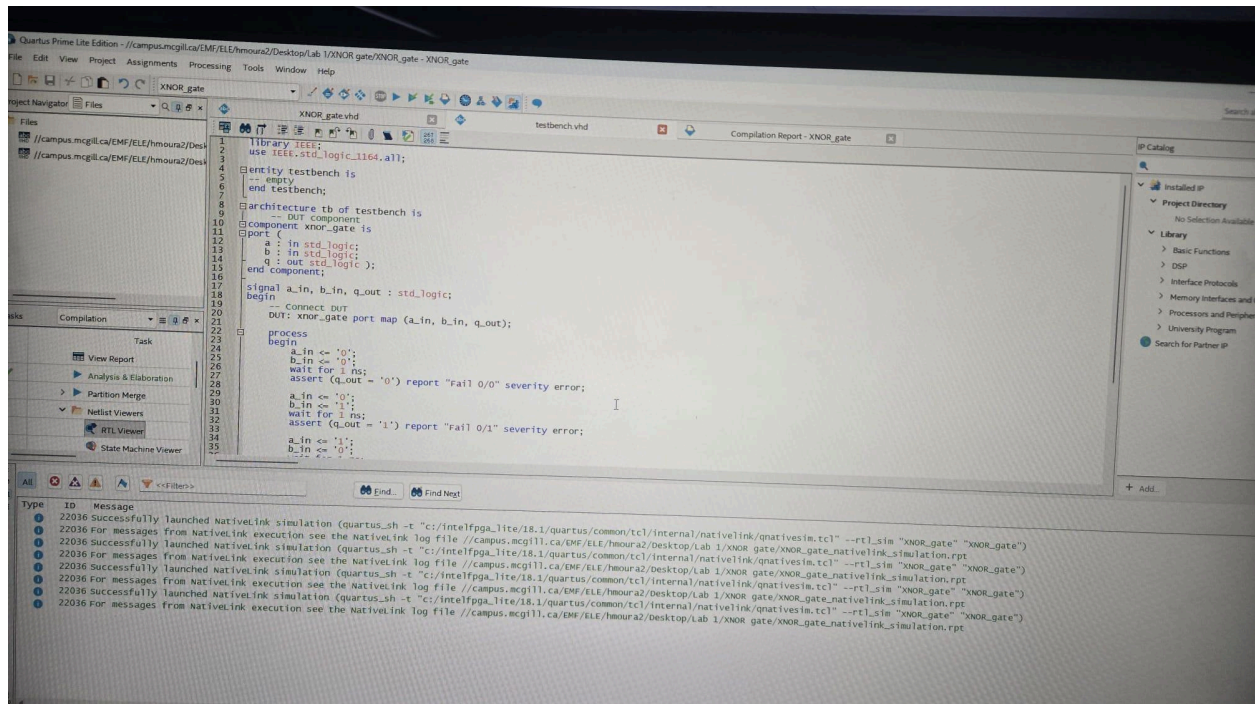
XNOR Gate VHDL Source



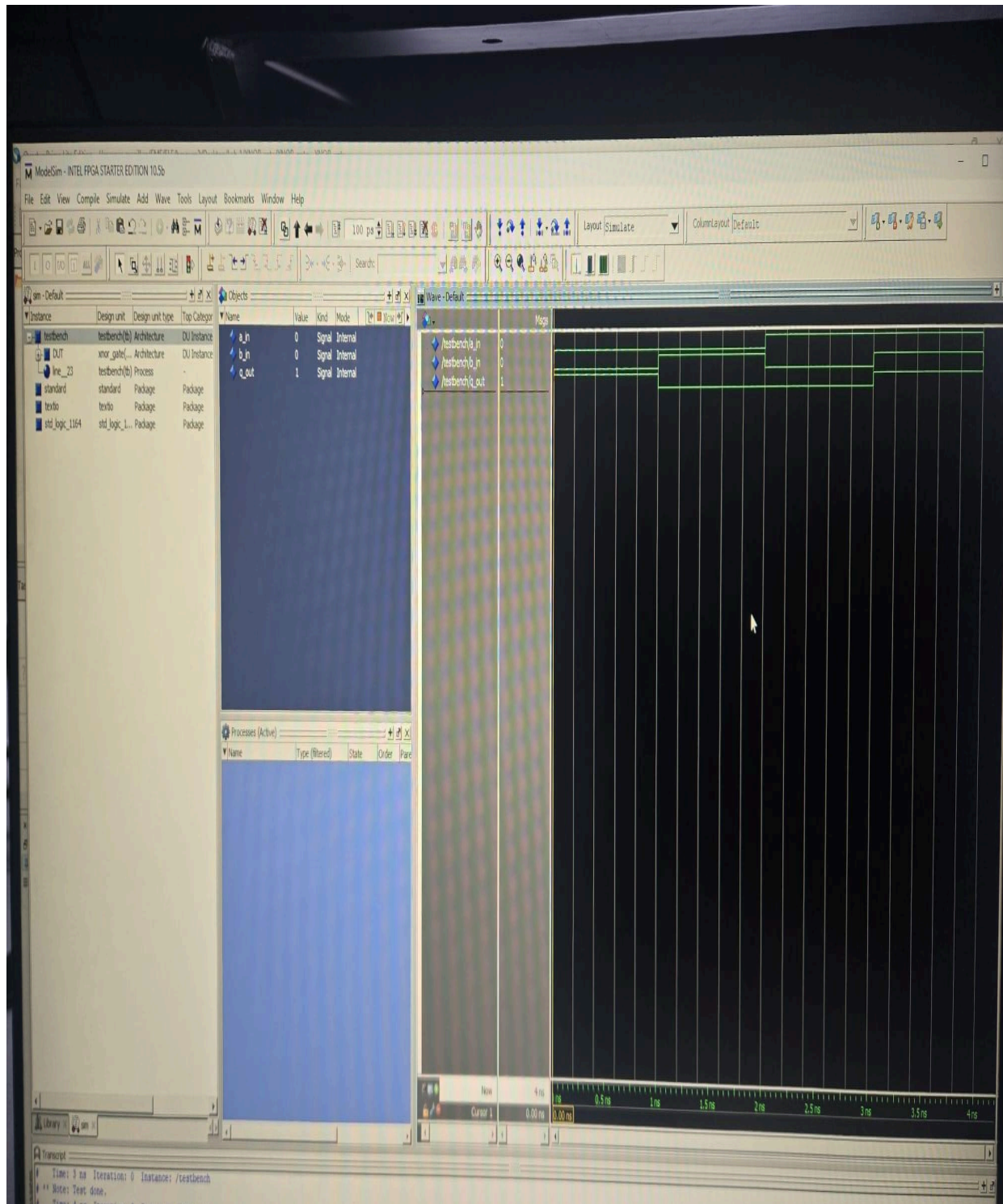
XNOR Gate RTL Schematic



Logic synthesis result: Quartus implements the design as a single 2-input XNOR gate



XNOR Gate testbench and Run logs



XNOR Gate Simulation Waveform (Results)

The output q is 1 whenever both inputs a and b are the same (both 0 or both 1) and 0 whenever the inputs are different (one 0 and one 1)

a = 0	a = 0	a = 1	a = 1
b = 0	b = 1	b = 0	b = 1
q = 1	q = 0	q = 0	q = 1

Conclusion:

This assignment provided us with valuable hands-on experience in VHDL programming and simulation using the Quartus CAD software.

We started by implementing and testing the OR gate, then proceeded to develop the XNOR circuit using two approaches: directly using the XNOR operator and by reconstructing it with basic gates such as AND and NOT. Through this, we successfully built and verified the functionality of each circuit.

Creating test benches, simulating the designs, and analyzing the RTL schematics helped us deepen our understanding of key concepts in digital design and simulation.

In summary, this assignment played a crucial role in improving our knowledge of VHDL syntax, simulation tools, and circuit validation techniques.

